

ADHITHYA A

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EDUCATION

SRM Institute of Science and Technology

Expected Graduation: 2029

Bachelor of Technology, Computer Science and Engineering

PROJECTS

Library Management System

2026

Python, Tkinter, SQLite, SQL

- Built a desktop application with a tabbed Tkinter GUI (Books, Members, Issue/Return) backed by a relational SQLite database.
- Designed a normalized three-table schema (books, members, issues) using foreign key relationships and JOIN queries to track live issue/return status.
- Implemented full CRUD functionality with parameterized SQL queries to prevent SQL injection.

Maze Generator & Solver

2026

Python, Pygame, Breadth-First Search, Recursion

- Developed a maze generator using recursive backtracking to procedurally create randomized, uniquely-solvable mazes.
- Implemented Breadth-First Search (BFS) graph traversal to compute and visualize the shortest path between start and end points.
- Built a real-time Pygame visualization with interactive maze regeneration on keypress.

Student Management System

2026

Python, Object-Oriented Programming, Tkinter, File I/O

- Built a console and GUI-based application to manage student records using object-oriented design principles.
- Implemented persistent data storage using file handling, enabling data to be saved and reloaded across sessions.

Personal Portfolio Website

2026

HTML, CSS, JavaScript

- Designed and built a fully responsive personal portfolio from scratch using semantic HTML and custom CSS, including a light/dark mode toggle.
- Deployed the site using GitHub and Render for public accessibility.

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS, JavaScript

Frameworks & Libraries: Tkinter, Pygame

Databases: SQLite

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Recursion, File I/O, Relational Database Design

Tools: Git, GitHub, VS Code